PLC ALARM / FAULT MATRIX

Alarm Code	Fault Condition	Detection State	PLC Action	Indication	Reset / Recovery
ALM-001	Safety Not Ready / E-Stop / Guard Open	Any State	Transition to S00; Process Outputs Off	P28 Red ON	Safety Restore; P00 ON; P07 ON
ALM-101	Transfer to Test Timeout	S30	P20 OFF; Transition to SF0	P28 Red ON; P29 Buzzer ON	P03 Reset + Safety Ready
ALM-102	Clamp Close Timeout	S50	P22 OFF; Transition to SF0	P28 Red ON; P29 Buzzer ON	P03 Reset + Safety Ready
ALM-103	Test Complete Timeout	S60	P22 OFF; D0022 = 0; Transition to SF0	P28 Red ON; P29 Buzzer ON	P03 Reset + Safety Ready
ALM-104	Invalid Test Result	S60	P22 OFF; D0022 = 0; Transition to SF0	P28 Red ON; P29 Buzzer ON	P03 Reset + Result Source Verified
ALM-105	Clamp Release Timeout	S70	P22 OFF; Transition to SF0	P28 Red ON; P29 Buzzer ON	P03 Reset + Safety Ready
ALM-106	Gate Position Timeout	S81-P/F/R	P23~P25 OFF; Transition to SF0	P28 Red ON; P29 Buzzer ON	P03 Reset + Gate Home
ALM-107	Outfeed Timeout	S90	P20 OFF; P23~P25 OFF; Transition to SF0	P28 Red ON; P29 Buzzer ON	P03 Reset + Outfeed Clear
ALM-108	D0023 Heartbeat Unchanged > 3 s MCU / Modbus Communication Fault	S30~S90 AUTO CYCLE ACTIVE	P20 OFF; P22~P25 OFF; Transition to SF0	P28 Red ON; P29 Buzzer ON	Communication Restored + P03 Reset

NOTE:
ALM-001 IS SAFETY STATUS / S00 CONDITION.
ALM-101 THROUGH ALM-108 ARE PLC PROCESS FAULTS / SF0 CONDITIONS.

PLC TIMER SETTING MATRIX

Timer	State	Function	Start Condition	Timeout Condition	Initial Setting	Final Setting
T0001	S30	Transfer to Test Timeout	P20 ON; P05 OFF	P05 not ON	10 s	Commissioning
T0002	S50	Clamp Close Timeout	P22 ON; P08 OFF	P08 not ON	3 s	Commissioning
T0003	S60	Test Complete Timeout	D0022 = 1; D0021 = 0	D0021 not ON	30 s	Commissioning
T0004	S70	Clamp Release Timeout	P22 OFF; P08 ON	P08 remains ON	3 s	Commissioning
T0005	S81-P/F/R	Gate Position Timeout	Selected Gate ON	P09/P0A/P0B not ON	3 s	Commissioning
T0006	S90	Outfeed Timeout	P20 ON; P06 OFF	P06 not ON	10 s	Commissioning
T0007	S30~S90	MCU Communication Watchdog	Auto Cycle Active	D0023 unchanged	3 s	Commissioning

NOTE:
ALL INITIAL TIMER VALUES ARE PROVISIONAL.
FINAL SETTINGS SHALL BE VERIFIED DURING COMMISSIONING.

TOWER LAMP / BUZZER STATUS MATRIX

PLC State	P26 Green	P27 Yellow	P28 Red	P29 Buzzer	Operator Meaning
S00 Safety Not Ready	OFF	OFF	ON	OFF	E-Stop / Door Open / Safety Not Ready
S10 Ready / Idle	ON	OFF	OFF	OFF	Safety Ready / Waiting for Start
S20 Wait Infeed	ON	OFF	OFF	OFF	Auto Mode / Waiting for Part
S30~S90 Auto Cycle	ON	OFF	OFF	OFF	Automatic Cycle Running
Controlled Normal Stop	OFF	ON	OFF	OFF	Controlled Stop / Operator Attention
Manual Mode	OFF	ON	OFF	OFF	Manual / Setup Mode
SF0 Fault Hold	OFF	OFF	ON	ON	Process Fault / Reset Required

NOTE:
P29 BUZZER MAY BE IMPLEMENTED AS CONTINUOUS OR PULSED ALARM.
FINAL AUDIBLE ALARM PATTERN SHALL BE DEFINED IN PLC LADDER LOGIC.

[GENERAL NOTES]

[1] MCU-01 SHALL UPDATE D0023 HEARTBEAT COUNTER AT 500 ms INTERVALS DURING NORMAL OPERATION.

[2] PLC-01 SHALL SET ALM-108 IF D0023 DOES NOT CHANGE FOR THE CONFIGURED WATCHDOG PERIOD DURING AUTO CYCLE.

PLC OUTPUT INTERLOCK AND MCU TEST HANDSHAKE:
SEE DWG 011 / SHEET 2 OF 2.

NO.	DATE	REVISION
A	2026-09-10	INITIAL RELEASE

DWG TITLE
EDS Wafer Map Linked Package
Final Test and Multi-Bin Sorting Cell
PLC ALARM / TIMER / INDICATOR MATRIX

DRAWN BY: J. H. KO
DATE: 2026-09-10
DWG NO: 012
SHEET NO: 1 OF 1